

AK VisionFlow — Phase 8

Claude Development Prompt
WhatsApp + Automation + Customer Portal

Critical context: Phase 1–7 provide the core POS, inventory, ERP/accounting, procurement, multi-branch, warehouse, optical, clinic, patient, prescription, lab and CRM foundation. Phase 8 is an incremental upgrade. Inspect the existing repository first. Do not rebuild working functionality, reset the database, or delete/corrupt existing tenant data.

1. Mission

Turn AK VisionFlow into an automated customer-communication and engagement platform. WhatsApp, notifications, reminders, workflow automation and a customer portal must be integrated with the existing sales, accounting, optical, clinic, appointments, lab, inventory and CRM systems rather than operating as disconnected tools.

2. Product Goal

Record the business event once → automatically trigger the right internal action, notification and customer communication.

Examples: optical job becomes Ready → notify customer; invoice becomes overdue → send payment reminder; appointment is tomorrow → send reminder; low stock → notify authorized staff; daily close → generate owner summary.

3. Non-Negotiable Rules

- Preserve all existing Phase 1–7 functionality.
- Never reset, truncate, recreate or destructively migrate the production database.
- All schema changes must use safe, versioned migrations.
- Existing customers, patients, suppliers, products, inventory, sales, purchases, accounting records, branches, optical orders, prescriptions, appointments and lab jobs must remain intact.
- Maintain strict tenant isolation for all communication, automation, templates, customer portal and notification records.
- Backend authorization is mandatory; UI-only restrictions are insufficient.
- Never send a message to the wrong tenant, customer or patient.
- Do not expose sensitive clinical information in messages unless explicitly configured and authorized.
- Automation must be idempotent and must not send duplicate messages after retries or sync events.
- Preserve offline/sync behavior and do not allow queued offline actions to bypass authorization.
- Respect provider rate limits, failures, opt-outs and delivery status.
- Run the full existing test suite and add comprehensive Phase 8 tests.

4. WhatsApp Integration Architecture

Implement a provider-agnostic messaging layer so AK VisionFlow is not permanently tied to one WhatsApp provider. The system should support a configurable WhatsApp Business/API provider and maintain message status internally.

- Tenant-level WhatsApp configuration.
- Provider credentials stored securely and never exposed to the frontend.
- Template management.
- Approved-template support where required by the provider.
- Message queue.
- Retry handling with exponential/backoff strategy where appropriate.

- Delivery status tracking.
- Failed-message tracking.
- Message history.
- Opt-out/consent handling.
- Rate-limit awareness.
- Webhook handling for delivery/read/failure events where supported.
- Idempotency keys to prevent duplicate sends.
- Provider abstraction so another provider can be added later.

5. WhatsApp Message Templates

- Invoice / Sale Receipt.
- Payment Receipt.
- Outstanding Payment Reminder.
- Customer Statement.
- Optical Order Confirmation.
- Optical Order Status Update.
- Optical Job Ready for Collection.
- Optical Order Delivered.
- Appointment Confirmation.
- Appointment Reminder.
- Appointment Cancellation/Reschedule.
- Follow-up Reminder.
- Prescription/clinical summary only where explicitly permitted and appropriately secured.
- General promotional/campaign message.
- Custom tenant-approved message templates.

6. Event-Driven Automation Engine

Create a reusable automation framework based on business events, conditions, actions and execution history.

- Event definitions: sale completed, payment received, invoice overdue, optical order created, job ready, job delayed, appointment booked, appointment approaching, stock low, expiry approaching, purchase approved, transfer completed, etc.
- Conditions: branch, customer segment, amount, due days, order type, status and other safe configurable fields.
- Actions: WhatsApp message, in-app notification, email where supported, dashboard alert, task/follow-up creation.
- Delay/schedule support for appropriate events.
- Automation enable/disable controls.
- Tenant-specific automation settings.
- Execution log.
- Success/failure status.
- Retry handling.
- Duplicate prevention.

- Permission-aware execution.
- Safe failure behavior: one failed notification must not roll back the underlying business transaction.

7. High-Value Default Automations

- Optical job marked READY → customer notification.
- Appointment scheduled → confirmation.
- Appointment approaching → reminder.
- Appointment no-show → follow-up task/message where configured.
- Invoice becomes overdue → payment reminder.
- Customer requests/receives statement → statement message.
- Low stock threshold reached → internal alert.
- Medicine expiry approaching → internal alert.
- Optical job delayed → internal alert + optional customer communication.
- Daily business closing → owner/admin summary notification.
- Customer inactive for configurable period → CRM follow-up opportunity.
- High-value/priority customer requires follow-up → internal reminder.

8. Customer Communication Center

- Unified message history per customer/patient where permitted.
- Search/filter by date, customer, branch, event, channel and status.
- Sent, delivered, read, failed and queued states where provider supports them.
- Retry failed messages safely.
- Manual resend with permission.
- Template preview.
- Variable/placeholder preview before sending.
- Message audit trail.
- Tenant-level communication statistics.
- Do not display sensitive clinical details to unauthorized staff.

9. Customer Portal

Provide customers/patients with a secure, mobile-friendly portal connected to their existing records.

- Secure customer login/verification.
- Profile and contact information.
- Appointments and appointment status.
- Appointment booking/request where enabled.
- Prescriptions and prescription history subject to authorization.
- Optical orders and current status.
- Invoices.
- Payments and receipts.
- Outstanding balance.

- Order history.
- Relevant follow-up information.
- Secure document access where supported.
- Logout/session management.
- Tenant-aware branding/configuration where appropriate.

10. Customer Portal Security

- Customers must only access their own records.
- Use secure authentication and session/token handling.
- Do not expose sequential IDs as the sole authorization mechanism.
- Backend must verify ownership on every portal API.
- Protect prescriptions and clinical information with additional authorization rules.
- Rate-limit login/verification attempts.
- Provide secure account recovery/verification mechanisms.
- Do not store unnecessary sensitive information in browser storage.
- Audit security-sensitive portal events.

11. Customer Self-Service

- Request/book appointment.
- View order status.
- View invoice/receipt.
- View outstanding amount.
- View permitted prescription history.
- Request follow-up.
- Request callback/contact.
- Update permitted profile fields.
- Communication preferences/opt-out.

12. CRM Automation

- Identify inactive customers.
- Identify repeat customers.
- Identify high-value customers.
- Prescription follow-up reminders.
- Optical replacement/follow-up opportunities based on configurable time windows.
- Outstanding-payment follow-ups.
- Customer segmentation.
- Campaign targeting by safe business criteria.
- Track campaign/message outcomes.
- Avoid spam: frequency limits, opt-out and tenant-configured communication rules.

13. Internal Notifications

- In-app notification center.
- Unread/read status.
- Low-stock notifications.
- Expiry alerts.
- Pending approval alerts.
- Delayed optical-job alerts.
- Overdue receivable alerts.
- Transfer alerts.
- Sync failure alerts.
- Critical system/security alerts.
- Role-aware notification delivery.

14. Daily Owner Summary

Integrate with the Advanced Business Command Center to generate a concise daily business summary for authorized owners/admins.

- Sales.
- Gross profit / profit where available.
- Collections.
- Expenses.
- Receivables/payables.
- Low stock.
- Expiry risks.
- Pending/delayed optical jobs.
- Appointments.
- Important approvals.
- Unusual activity or operational warnings.
- Link back to the relevant dashboard/report.

15. Communication Preferences

- Tenant-level enable/disable controls.
- Customer communication opt-in/opt-out.
- Per-message-type preferences where practical.
- Business-hours rules.
- Maximum message frequency for campaigns.
- Template language/configuration.
- Branch-specific sender/configuration where provider supports it.

16. Audit, Compliance & Safety

- Record who triggered manual messages.

- Record automation execution.
- Record template used.
- Record recipient/customer reference.
- Record provider message ID where available.
- Record delivery/failure status.
- Never log provider secrets.
- Do not put sensitive clinical information into generic application logs.
- Maintain a clear trail for customer communication.

17. Offline & Sync Requirements

- Business transactions completed offline must remain the source event for automation after successful synchronization.
- Do not send duplicate WhatsApp messages because the same transaction syncs more than once.
- Use event IDs/idempotency keys.
- Queue external communication safely after the underlying transaction is confirmed.
- A messaging-provider outage must not corrupt sales, payments, optical jobs or appointments.
- Show communication queue/failure state to authorized staff.

18. Advanced Command Center Integration

- Messages sent/delivered/failed.
- Pending automation jobs.
- Failed automations requiring attention.
- Overdue customer follow-ups.
- Upcoming appointments.
- Pending/delayed optical jobs.
- Customer engagement indicators.
- Portal activity where appropriate.
- Communication performance by branch.
- Drill-down to communication history.

19. Reports & Analytics

- Communication volume by channel/status.
- WhatsApp delivery/failure report.
- Automation execution report.
- Customer follow-up report.
- Appointment reminder report.
- Payment reminder report.
- Optical order notification report.
- Customer engagement/response indicators where available.
- Campaign performance report where campaigns are implemented.
- Branch-wise communication activity.

20. UX Requirements

- Keep communication setup simple for small shops.
- Provide sensible default automations.
- Use clear status badges: queued, sent, delivered, read, failed.
- Provide safe preview before manual or campaign sending.
- Make portal fully responsive/mobile-first.
- Maintain existing light/dark mode and design language.
- Use clear loading, empty and error states.
- Do not overwhelm shop users with technical provider details.

21. Testing Requirements

- Test tenant isolation for messages, templates, automations and portal data.
- Test customer ownership checks on every portal endpoint.
- Test duplicate-prevention/idempotency for message events.
- Test provider failure and retry behavior.
- Test webhook validation and status updates where applicable.
- Test appointment, payment and optical-job automations.
- Test opt-out and communication preferences.
- Test that sensitive clinical data is not exposed to unauthorized users.
- Test offline transaction → sync → single automation execution.
- Test RBAC for manual messaging and automation configuration.
- Run the complete regression suite from previous phases.

22. Migration & Data Safety

Inspect existing customer/patient, appointment, optical-order, payment and CRM schemas before adding communication and portal entities. Reuse existing IDs and relationships. Add only the minimum new tables/fields needed for message logs, templates, automation rules, events, portal identities and preferences. Do not duplicate customer/patient master data.

23. Required Deliverables

- Provider-agnostic WhatsApp integration layer.
- WhatsApp templates and secure configuration.
- Message queue/retry/idempotency system.
- Communication history and status tracking.
- Event-driven automation engine.
- High-value default automations.
- Customer Communication Center.
- Secure Customer Portal.
- Customer self-service features.
- CRM follow-up automation.
- Internal notification center.

- Daily owner summary integration.
- Communication preferences and opt-out controls.
- Audit/security controls.
- Command Center integration.
- Reports/analytics.
- Safe migrations, automated tests and documentation.

24. Acceptance Criteria

- Existing Phase 1–7 functionality remains intact.
- No existing data is deleted, reset or corrupted.
- WhatsApp configuration is secure and tenant-isolated.
- Messages are queued, tracked and retried safely.
- Duplicate messages are prevented.
- Core business transactions are not rolled back because messaging fails.
- Appointment, payment and optical-job notifications work correctly.
- Customer portal users can access only their own permitted information.
- Sensitive clinical information remains protected.
- Automation rules are auditable and permission-aware.
- Opt-out/preferences are respected.
- Owner/admin can see communication and automation health from the Command Center.
- All existing tests plus new Phase 8 tests pass.
- No critical security, privacy or data-integrity issue remains open.

25. Claude Execution Order

1. Inspect the existing repository, CRM, patient, appointment, optical, payment and Command Center architecture.
2. Run the full existing test suite and establish a baseline.
3. Design the communication/event/automation/portal data model.
4. Design safe migrations and test them against realistic existing data.
5. Implement provider abstraction and secure WhatsApp configuration.
6. Implement templates, message queue, retries, status tracking and idempotency.
7. Implement event-driven automation engine.
8. Implement high-value default automations.
9. Implement Communication Center and internal notifications.
10. Implement secure Customer Portal and self-service.
11. Implement CRM follow-up automation and communication preferences.
12. Integrate daily owner summary and Command Center metrics.
13. Add reports and audit controls.
14. Add comprehensive automated tests.
15. Run full regression, security, privacy and data-integrity review.

16. Document configuration, provider setup, migrations, operational procedures and known limitations.

Final Instruction to Claude

Build Phase 8 as a safe, production-quality incremental upgrade. Do not rebuild the existing core and do not reset the database. The goal is to make AK VisionFlow proactive: when something happens in the business, the system should automatically perform the appropriate internal workflow and customer communication.

WhatsApp must be provider-agnostic, secure, auditable, retry-safe and idempotent. The Customer Portal must be secure and ownership-checked at the backend. Do not expose sensitive clinical information through generic messages or unauthorized portal access. Preserve existing accounting, inventory, multi-branch, optical, clinic, CRM, offline/sync and Command Center functionality. Keep the UX simple enough for small shops while making the automation powerful enough for larger businesses. The result must be tested, maintainable and ready for Phase 9: AI + Business Intelligence + Predictive Analytics.